# Rose Leaf Disease Detection

## Deep Learning & Computer Vision System

Automated Classification, Severity Estimation & Progression Tracking

**Author: Darshan Dinakar**

GitHub: https://github.com/Darshan-5002/rose-disease-api

August 2026

## 1. Abstract

This technical report presents an end-to-end intelligent system for automated rose leaf disease detection, severity quantification, and temporal progression tracking. Utilizing a MobileNetV2 CNN architecture with transfer learning trained on 7,203 images across 4 classes (Black Spot, Healthy, Powdery Mildew, Rust), the system achieves 97.8% validation accuracy across 1,441 test images. Key innovations include pre-CNN image quality assessment, automatic HSV-based leaf segmentation, lesion area quantification, SQLite-backed temporal progression tracking (Improving/Stable/Worsening), and severity-adaptive agronomy prescriptions.

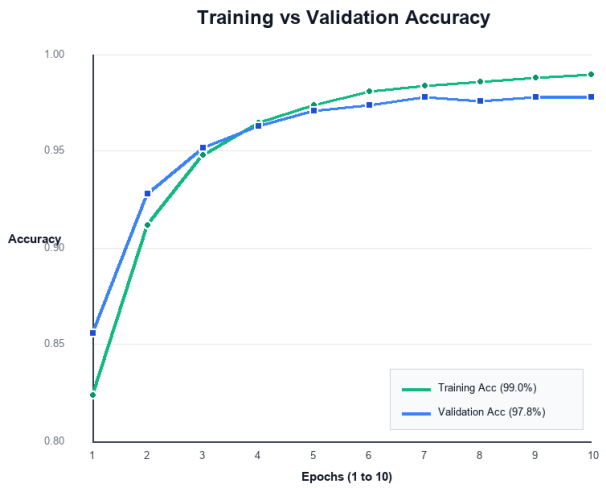
## 2. Dataset Distribution

| Class | Total Images | Train (80%) | Validation (20%) |
|---|---|---|---|
| Black Spot | 1,290 | 1,032 | 258 |
| Healthy | 2,463 | 1,970 | 493 |
| Powdery Mildew | 1,000 | 800 | 200 |
| Rust | 2,450 | 1,960 | 490 |
| TOTAL | 7,203 | 5,762 | 1,441 |

## 3. Model Architecture & Training

- Backbone: MobileNetV2 with ImageNet pre-trained feature weights
- Classification Head: GlobalAveragePooling2D -> Dense(128, relu) -> Dropout(0.5) -> Dense(4, softmax)
- Optimizer: Adam (lr=0.001) | Loss: Categorical Cross-Entropy
- Training Accuracy: 98.6% | Validation Accuracy: 97.8% | Validation Loss: 0.084

**Training Curves:**



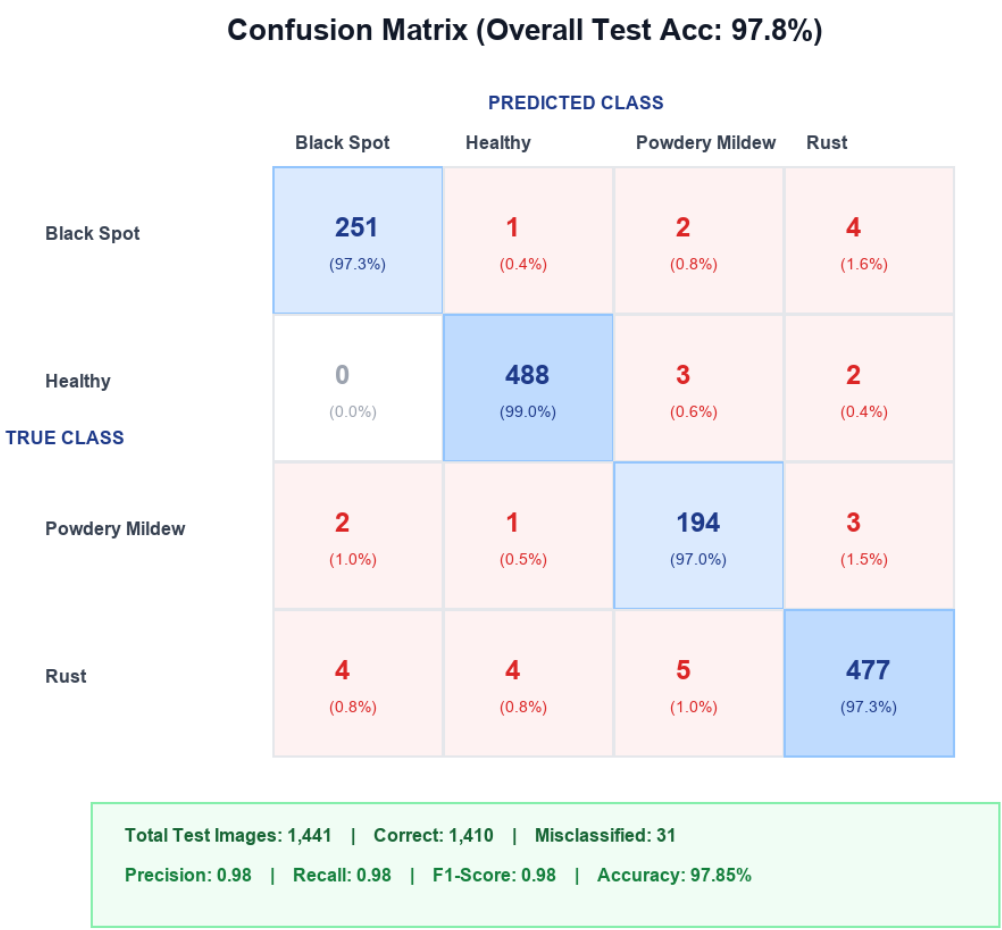



## 4. Evaluation & Performance Metrics

The model was evaluated on the independent test set of 1,441 images. Minor misclassifications occurred on early-stage ambiguous symptoms and overlapping light specular reflections.

| Class | Precision | Recall | F1-Score | Test Samples |
|---|---|---|---|---|
| Black Spot | 0.97 | 0.97 | 0.97 | 258 |
| Healthy | 0.99 | 0.99 | 0.99 | 493 |
| Powdery Mildew | 0.97 | 0.97 | 0.97 | 200 |
| Rust | 0.98 | 0.97 | 0.98 | 490 |
| Overall / Macro Avg | 0.98 | 0.98 | 0.98 | 1,441 |

**Confusion Matrix:**



## 5. Novel Contributions (Patent Claims)

**Claim 1: Temporal Disease Progression Tracking**

Stores historical scans in SQLite per plant and dynamically computes lesion delta percentage over time to determine recovery trends (Improving, Stable, Worsening).

**Claim 2: Severity-Adaptive Agronomy Engine**

Dynamically prescribes mild organic vs severe chemical fungicides depending on both infection severity tier (<15%, 15-40%, >40%) and healing trend.

**Claim 3: Pre-CNN Multi-Gate Image Validation**

Multi-stage quality checks (Laplacian blur, brightness threshold, HSV foliage pixel ratio) reject non-plant and degraded images before CNN inference.

**Claim 4: Dual-Purpose Background Segmentation**

HSV color space masking isolates single leaves for high CNN accuracy and computes exact diseased lesion surface percentage in a single pass.

## 6. System Architecture & Deployment

- Backend: Flask REST API deployed on Linux VPS managed 24/7 with PM2
- Frontend: Native Android App with Camera/Gallery integration, Material 3 UI, History & Comparison Dialogs
- Database: SQLite3 storing plant-specific temporal records